

Polaris Energy and Fuel Resilience for Remote Microgrids

Prepared by Team CarbonIQ
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Polaris helps operators use available renewable energy, batteries and diesel efficiently while planning around the next fuel delivery. The proposed prototype combines seven-day forecasts, constrained dispatch planning and fuel-risk recommendations.

The project in simple words

Imagine a research station or community facility far from the electricity grid. Sunlight and wind change, batteries have limited storage, and diesel must last until another delivery. Polaris helps the operator answer three practical questions:

Question	What Polaris proposes to show
What electricity will we need?	Hourly demand and renewable forecasts for the next seven days.
Which energy source should we use?	A feasible plan for renewables, battery charging or discharge, and diesel generation.
Will fuel last until delivery?	An estimated runway range, a shortfall warning and comparisons of possible actions.

A running example

Fuel is expected to last 20 days, but delivery is scheduled for day 25. Polaris tests whether moving flexible tasks to sunny hours, changing battery reserves or reducing nonessential use could close that five-day gap.

What the final-round prototype will demonstrate

One configurable engine will support a polar research profile and a remote community profile. Operators will compare ordinary operating rules with Polaris, inspect the reasons for recommendations, and see the effect on fuel, emissions and essential-service availability.

Evidence status: This document specifies a proposed prototype. The running example and savings figures are illustrative. No station deployment or measured Polaris performance is claimed.

The problem and intended users

Proposal question 22 | Problem statement and operating context

Problem statement. Remote microgrid operators need to coordinate electricity demand, variable renewable generation, battery reserves and a finite fuel supply under uncertain weather and resupply conditions. Poor coordination can increase fuel consumption, waste renewable energy or expose essential services to shortages.

A fuel-level reading alone cannot tell an operator whether the facility will remain powered until delivery. The missing decision is how to operate today while preserving enough energy and fuel for the days ahead.

Two applications of the same planning engine

Profile	Essential services	Flexible activities to schedule
Polar research station	Configured communications, safety systems, heating support and essential instruments.	Selected laboratory tasks, noncritical equipment and other activities the operator marks as flexible.
Remote community facility	Configured clinic equipment, communications, essential lighting and priority water services.	Pumping into available storage, workshop activity and other approved tasks that can move within a deadline.

Facts and modeling assumptions

Maitri and Bharati are Indian Antarctic research stations. Himadri is at Ny-Ålesund in the Arctic; Kings Bay supplies shared electricity infrastructure there. Their actual equipment and operating arrangements must be verified separately. The prototype will not describe all three as identical standalone solar-battery-diesel systems. [5, 6]

Prototype equipment capacities, demand patterns, generator fuel curves, battery limits, fuel inventory and delivery dates will be explicit configuration inputs. Weather will come from real historical records; demand and equipment behavior will be labelled simulated wherever operational data is unavailable.

Why the five-day gap matters

The example compares a 20-day runway with a day-25 delivery. It does not assume a solution exists: Polaris must also show when the installed equipment and approved operating changes cannot close the gap.

What success means

Success is a reproducible reduction in fuel and direct combustion emissions, with essential-service performance reported alongside savings. Deferred work, nonessential curtailment, unmet demand and remaining risk must stay visible. A lower fuel bill alone is not enough evidence.

How Polaris works

Proposal question 23 | Forecasting, constrained dispatch and fuel-risk assessment

Polaris will turn weather history and operating data into forecasts, then use those forecasts to plan energy dispatch. Fuel-risk analysis will compare consumption scenarios with the expected resupply date. New readings will update the plan.

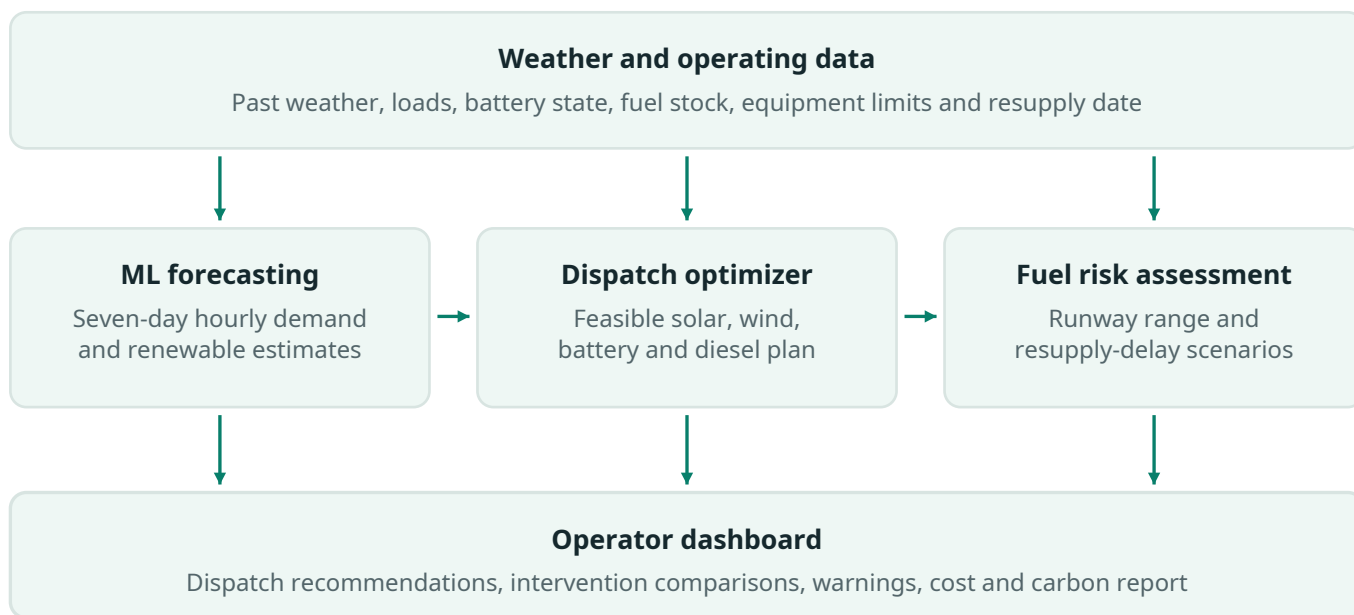


Figure 1. Shared inputs feed forecasting, equipment constraints and fuel-risk analysis. The forecast informs dispatch; the resulting plan informs fuel-runway scenarios.

Forecast horizon and planning cycle

XGBoost or LightGBM models will estimate 168 hourly demand and renewable values. Uncertainty bands will be evaluated on separate calibration data. The optimizer will replan hourly in simulation, implement the next action and update the battery and fuel state. Estimates beyond seven days will use declared seasonal and stress scenarios, not claim a precise 25-day weather forecast. [2]

Mode	Priority	What remains protected
Economy	Reduce operating cost, including fuel, generator starts and battery wear.	Equipment limits and configured essential loads.
Green	Reduce direct diesel combustion CO ₂ , subject to operating and wear limits.	The same service and physical constraints.
Survival	Preserve fuel and reserves toward the resupply date under conservative scenarios.	Essential loads; approved flexible tasks may move.

Physical checks: energy balance, generator limits, fuel use, battery efficiency and state of charge; prevent simultaneous battery charging and discharge. If essential demand cannot be served, display the unmet demand and infeasibility clearly.

Innovation and advanced features

Proposal question 24 | A resupply-aware operator workflow

The proposed distinction is the connection between day-to-day energy dispatch and the next fuel delivery. Polaris will explain how weather uncertainty, delivery delays and operator-approved changes alter the risk of a shortfall.

Capability	Operator benefit	Evidence to demonstrate
Uncertainty-aware forecasts	See a plausible range instead of a single optimistic estimate.	Forecast errors and interval coverage on unseen data.
Resupply-aware planning	Understand whether reserves are adequate for delivery delays.	Normal-delivery and delayed-delivery scenarios.
Essential-load priorities	Keep the consequence of a recommendation visible.	Served and unmet essential demand alongside fuel savings.
Intervention comparisons	Choose between actions using quantified estimates.	Fuel, cost, runway and service effects for each action.
Offline-capable operation	Review a cached simulation and recommendations without a live API.	Repeatable local demonstration and exported outputs.

Recommendations the operator can inspect

For each approved intervention, rerun the same scenario and show the change relative to the current plan. Examples include moving flexible tasks into renewable-rich hours, changing a battery reserve policy, or reducing selected nonessential demand. Essential loads are not silently downgraded.

The example as a decision

For the 20-day runway and day-25 delivery, each permitted action will be compared with the current schedule. Any additional runway will be reported from the simulation, together with deferred work and remaining shortfall risk.

Positioning against existing work

REopt supports energy-system dispatch optimization and resilience analysis, providing a relevant reference for evaluation. Polaris proposes a configurable, explainable workflow linking rolling forecasts, resupply scenarios and intervention decisions for remote operators. Its distinct contribution will be assessed through reproducible comparisons with existing approaches. [4]

Economy and Green may produce the same schedule when diesel dominates both cost and emissions. Their comparison will show the actual cost, carbon and battery-wear trade-offs.

Prototype and validation

Proposal question 25 | What the team will build and how results will be checked

The prototype will reuse one Python engine across two configurable profiles. A Streamlit dashboard will connect trained forecasts, an OR-Tools MILP planner, fuel-risk scenarios and transparent impact reporting.

Deliverable	Evidence supplied with the submission
Weather and profile data	Cached historical weather, timestamps, provenance and clear labels for simulated demand and equipment.
Trained model repository	Model artifacts, feature definitions, training scripts, chronological data splits and inference examples.
Dispatch and fuel-risk engine	Three operating modes, constraint checks, resupply scenarios and intervention comparisons.
Dashboard and annual evaluation	Two profile presets, a seven-day replay and a held-out one-year simulation report.
Reproducible package	Dependency versions, fixed random seeds, run instructions and exportable forecasts, dispatch and carbon calculations.

A fair comparison

Use the same hardware, demand, actual weather, fuel supply and starting battery state for all runs. Compare **A**: rule-based operation; **B**: optimization using simple persistence or seasonal forecasts; and **C**: Polaris using trained forecasts. This separates the value of optimization from the added value of ML.

Train and calibrate on earlier periods, then test on a separate full year. At each simulated decision, use only information available at that time. Future observed weather is for scoring realized outcomes, not an input to the decision. ERA5 is historical reanalysis; an operational deployment would need a weather-forecast source. [2]

Acceptance checks before the jury demo

- Compare forecast MAE and RMSE with simple baselines; report results and uncertainty coverage across seasons and stress periods.
- Test renewable shortages, winter low-solar periods, delayed delivery, low battery reserves, missing inputs and infeasible supply.
- Verify energy balance, battery limits and nonnegative fuel; expose solver failures and unmet demand.
- Report curtailed or deferred demand with fuel and carbon savings; replay both profiles from the documented inputs.

Submission: 8 October 2026, 5-6 PM IST. **Jury pitch and demo:** 9 October, 9 AM-noon. The work will be organized around these published deadlines. [7]

Impact and transparent calculations

Proposal question 26 | Environmental benefit with clearly stated assumptions

Polaris aims to reduce unnecessary diesel use while making service reliability and fuel-shortfall risk visible. Impact will come from simulation results; reductions are not guaranteed in every season or equipment configuration.

Illustrative annual example

The following inputs explain the calculation method. They are assumptions, not verified consumption at a named station or measured Polaris results.

Quantity	Calculation or assumption	Illustrative value
Baseline consumption	Assumed annual consumption	84,000 litres/year
Diesel saved	Assumed saving to illustrate impact	22,500 litres/year
Fuel reduction	22,500 divided by 84,000	26.8%
Fuel expenditure avoided	22,500 × assumed ₹80/litre	₹18 lakh/year
Direct CO ₂ avoided	22,500 × 2.70 kg, divided by 1,000	60.75 tonnes/year approximately 61 tonnes

The rounded factor of 2.70 kg CO₂/litre is consistent with the EPA value of 10.21 kg CO₂ per US gallon for distillate fuel oil No. 2, divided by 3.78541 litres per US gallon. Actual fuel properties can differ. The example counts direct combustion CO₂ only; it excludes fuel delivery, upstream emissions, equipment lifecycle emissions and other greenhouse gases. [3]

Metrics the evaluation must report

Fuel and money: annual litres used and saved, percentage reduction, assumed unit price and fuel expenditure. **Energy:** available and used renewable kWh, renewable utilization, curtailment, battery throughput and generator starts. **Service:** essential demand served, unmet kWh and flexible work deferred or curtailed. **Resilience:** estimated runway range, remaining reserves and delivery-delay shortfalls.

What the runway example establishes

The 20-day runway and day-25 delivery demonstrate a planning problem. Whether an intervention closes the gap depends on available generation, storage, demand flexibility and the weather scenario.

Scaling and sustainability alignment

Deployment potential will be estimated from documented site profiles and explicit adoption assumptions. The project aligns with clean energy, resilient infrastructure, community services, resource efficiency and climate action, supporting SDGs 7, 9, 11, 12 and 13.

Track 3 alignment and pilot pathway

Required evidence, judging criteria, references and a short glossary

Track 3 requirement	Submission evidence
Trained ML repository	Versioned models, training scripts, data provenance and run instructions.
Sample inference outputs	Seven-day demand and renewable predictions with evaluation examples.
Carbon-abatement projections	Baseline comparison, fuel differences, factors and stated boundaries.

Judging alignment [1]: Innovation and originality (25%) - resupply scenarios and intervention comparisons. Technical feasibility (25%) - working code and constraint checks. Environmental impact (25%) - measured simulation savings with service outcomes. Presentation and roadmap (25%) - clear operator demo and a credible pilot pathway.

A realistic progression

Simulation: complete reproducible evaluation and stress tests. **Operational-data validation:** seek a willing operator, verify equipment and fuel data, and backtest with their permission. **Supervised pilot:** run recommendations in advisory or shadow mode, track outcomes and refine constraints before considering equipment control. No partnerships or deployments are presumed.

Short glossary

Microgrid: a local electricity system. **Dispatch:** deciding which source supplies power at each time. **MILP:** mathematical planning with numerical and on/off decisions. **SOC:** battery state of charge. **Fuel runway:** estimated time before usable fuel is exhausted under stated conditions. **Held-out evaluation:** testing on data kept separate from training.

Sources and linked references

[1] [Greenovators Track 3 themes and evaluation rules](#)

Required deliverables and four equally weighted judging criteria.

[2] [ECMWF Complete ERA5 global atmospheric reanalysis](#)

Historical weather data; reanalysis is distinct from an operational forecast.

[3] [US EPA 2025 GHG Emission Factors Hub](#)

Table 1 stationary combustion; fuel factor and combustion-only boundary.

[4] [National Laboratory of the Rockies What Is REopt](#)

Existing optimization and resilience capabilities.

[5] [NCPOR Indian Arctic programme and Himadri](#)

Himadri location and the shared research-base context.

[6] [Kings Bay Strategy 2024 to 2027](#)

Shared electricity infrastructure at Ny-Ålesund.

[7] Supplied local documents: *Hackathon Program Schedule.pdf* and *Brochure Greenovators hackathon.pdf*. Original concept: *Polaris_CarbonIQ.pdf* and Team CarbonIQ's updated brief.

References reviewed 5 October 2026. This proposal documents planned capabilities; measured results should be added after the prototype evaluation.